Course notes

Differentiating a composite

If u is differentiable on I and v differentiable on an interval containing $u(I)$, then $(v \circ u)' = (v' \circ u)u'$. The domains must be checked before calculating.

$$(u^n)' = nu'u^{n-1}, \quad (e^u)' = u'e^u, \quad (\frac{1}{u})' = -\frac{u'}{u^2}, \quad (\sqrt{u})' = \frac{u'}{2\sqrt{u}} \quad (u > 0).$$

For example, $f(x) = \sqrt{3x+1}$ is differentiable for $x > -\frac{1}{3}$ & $f'(x) = \frac{3}{2\sqrt{3x+1}}$. The sign of f' determines the variations of f; the sign of f does not determine them.

Convexity

A function f is convex on an interval I if its curve lies below its chords: for $x, y \in I$ and $t \in [0,1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

For a twice-differentiable function, we accept the equivalences: f convex; f' increasing; $f'' \geq 0$; the curve lies above all its tangents. The inequalities are reversed for a concave function. An inflection point is a point where the curve changes convexity.

Proof: position relative to the tangents. Suppose $f'' \geq 0$. Fix a and let $g(x) = f(x) - f(a) - f'(a)(x-a)$. Since f' is increasing, $g'(x) = f'(x) - f'(a) \leq 0$ for $x \leq a$ and ≥ 0 for $x \geq a$. Thus g reaches its minimum $g(a) = 0$ at a, therefore $f(x) \geq f(a) + f'(a)(x-a)$.

Examples. x^4 is convex and $f''(0) = 0$, but 0 is not the x-coordinate of an inflection point. x^3 changes convexity at 0. The convexity of the exponential function and its tangent at 0 give $e^x \geq 1 + x$.

Continuity and equations

f is continuous at a if $\lim_{x \to a} f(x) = f(a)$. Every differentiable function is continuous. The converse is false: $|x|$ is continuous at 0 but not differentiable at 0.

Intermediate value theorem

If f is continuous on $[a,b]$, any real between $f(a)$ and $f(b)$ has at least one preimage in $[a,b]$. If f is moreover strictly monotonic, this preimage is unique. We can also work on an open interval using the limits at the endpoints.

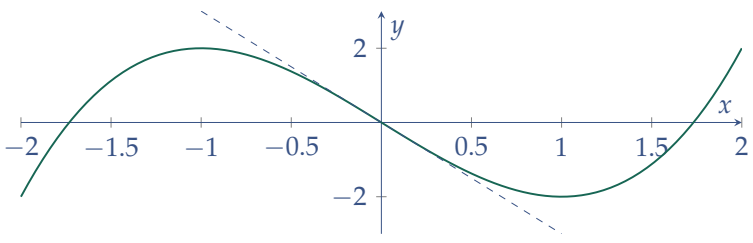
Worked example. $f(x) = x^3 + x - 1$ is continuous on $[0,1]$. Since $f(0) = -1$ and $f(1) = 1$, the IVT gives a zero in $]0,1[$. Since $f'(x) = 3x^2 + 1 > 0$, it is unique. $f(0{,}68) < 0 < f(0{,}69)$ provides bounds, without replacing the justifications of existence and uniqueness.

A piecewise-defined function. For $x \neq 1$, set $g(x) = \frac{x^2-1}{x-1}$, and set $g(1) = a$. Since $g(x) = x + 1$ when $x \neq 1$, $\lim_{x \to 1} g(x) = 2$. The function is continuous at 1 if and only if $a = 2$. The value at the point must therefore be compared with the limit, not replaced by it without justification.

Sequences and continuity. If $u_n \to \ell$ & f is continuous at ℓ, then $f(u_n) \to f(\ell)$. For a recursively defined sequence convergent $u_{n+1} = f(u_n)$, we get $\ell = f(\ell)$.

Bisection

For f continuous and $f(a)f(b) < 0$, calculate $m = \frac{a+b}{2}$. If $f(m) = 0$, stop; otherwise keep the half whose the endpoints have images of opposite signs. After n steps, the length is $\frac{b-}{2^n}$. The midpoint gives an error of at most half the length. The method does not guarantee uniqueness; that is proved separately.



$y = x^3 - 3x$ and its tangent at the inflection point.

Linking calculation, curve and argument

4.1 The inner factor and the domain

First use: S29.

Worked example · A two-step composition

For $f(x) = \sqrt{1+x^2}$, the argument is always strictly positive: f is differentiable on $\mathbb{R}$. The outer function has derivative $t \mapsto 1/(2\sqrt{t})$, the inner function has derivative $2x$. Their product gives $f'(x) = x/\sqrt{1+x^2}$. The fraction is negative for $x < 0$, positive for $x > 0$: f decreases then increases.

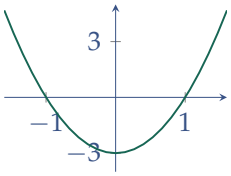
A1 · Immediate application Determine the domain of differentiability and the derivative of $g(x) = \sqrt{2x-1}$ and then $h(x) = e^{1-x^2}$. Check the sign of g'.

A2 · Fix a derivative "For $f(x) = (3x+1)^4$, we have $f'(x) = 4(3x+1)^3$." Explain the missing factor and suggest a check by expanding or using the rate of change at 0.

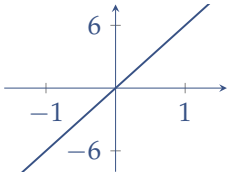
4.2 Match f, f' & f''

First use: S29 and S31.

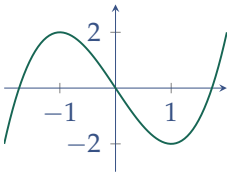
Curve A



Curve B



Curve C



Worked example · A positive decreasing function

For $g(x) = e^{-x}$, g is positive, $g' = -e^{-x} < 0$ and $g'' = e^{-x} > 0$. The curve goes down, but its negative slopes increase towards 0: it is convex. The three pieces of information are consistent.

A3 · Read before calculating The three curves correspond, in an order to be determined, to $f(x) = x^3 - 3x$, f' & f''. Match them using zeros, extrema and changes of convexity; then check by differentiating.

Property · Two readings not to be confused

$f'(a) > 0$ describes a positive slope of f, not a positive y-coordinate. $f''(a) > 0$ indicates locally that the slopes increase; this does not mean that f increases. For example e^{-x} decreases and remains convex.

4.3 Between a tangent and a chord

First use: S31; revisited in S33.

Worked example · Two bounds for x^2

On $[0;2]$, the chord joining $(0;0)$ to $(2;4)$ has equation $y = 2x$; the tangent at 1 has equation $y = 2x - 1$. Convexity gives 2 $x - 1 \leq x^2 \leq 2x$. The differences check out: $x^2 - (2x-1) = (x-1)^2 \geq 0$ and $2x - x^2 = x(2-x) \geq 0$ on this interval.

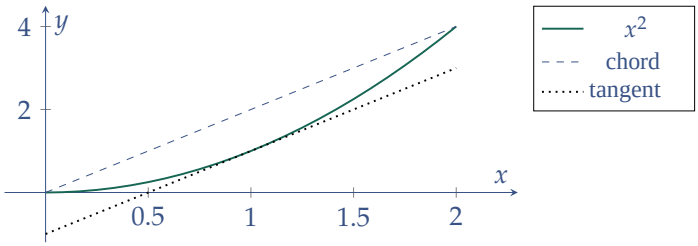


Figure 4.1. Compare the three y-values for $x = \frac{1}{2}$, then justify for every $[0;2]$.

A4 · Compare three functions For x^4, x^3 & $|x|$, study at 0 the continuity, differentiability and change of convexity. For both polynomials, explain why a vanishing second derivative is not enough to conclude.

4.4 Existence and uniqueness are two conclusions

First use: S32 and S33.

Worked example · Writing up an approximate solution

Let $F(x) = x^3 + x - 1$. It is a polynomial, so it is continuous. $F(0) < 0 < F(1)$ gives a solution in $]0;1[$ by the IVT. $F'(x) = 3x^2 + 1 > 0$ on $\mathbb{R}$ ensures that there cannot be two. The calculation $F(0{,}68) < 0 < F(0{,}69)$ locates this solution, but replaces neither existence nor uniqueness.

A5 · Two counterexamples Does the fn equal to -1 for $x < 0$ and 1 for $x \geq 0$ change sign without vanishing? Which hypothesis of the IVT is missing? For $x^2 = 1$, which hypothesis is missing to claim a unique solution on $\mathbb{R}$?

4.5 Keep information at every step

First use: S33 and S40.

Property · Bisection invariant

For a continuous function, maintaining $F(a)F(b) \leq 0$ preserves at least one zero in $[a;b]$. À At each step, the length is halved. The algorithm gives bounds; uniqueness must be established separately.

Worked example · A first interval

For $F(x) = x^2 - 2$ on [1;2], $F(1) = -1$, $F(2) = 2$. At the midpoint $m = 3/2$, $F(m) = 1/4 > 0$; the half that keeps the change of sign is $[1;3/2]$. Its width is 1/2.

A6 · Choice of strategy Prove that $e^{-x} = x$ has a unique solution in $[0;1]$. Perform two bisection steps, then determine how many steps guarantee an interval width of at most 1/128.

A first step: Reduce the equation to finding a zero of a function. The number of steps comes from the width, not of the values of the function.

One minute, spoken aloud. State the existence of a solution in one sentence containing the interval, the continuity and the values at the endpoints; state uniqueness in a separate sentence.

Delayed review · S47 then S66. Apply the same arguments to a logarithmic function, then to an area dependent on a bound.

Investigation

Derivatives and reading graphs - 35 min.

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. Let $\mathcal{C}_f$ its graph.

1. Calculate $f'(x)$ and $f''(x)$, then draw their sign tables.
2. Deduce the variations of f and its intervals of convexity and concavity.
3. Determine the coordinates of the inflection point of $\mathcal{C}_f$ and an equation of the tangent at this point.
4. Using the limits, the extrema and the intermediate value theorem, determine the number of real solutions of $f(x) = 1$.
5. Explain, from this example, why continuity is not enough to ensure the uniqueness of a solution.